Advanced Risk Modeling, Claims & Coverage Extensions: ICICI Lombard Motor Insurance

This research document analyzes the technical and digital frameworks governing ICICI Lombard's motor insurance portfolio, specifically focusing on usage-based underwriting, AI-integrated claims, and liability logic.

1. Advanced Risk Modeling: Usage-Based Insurance (UBI)

ICICI Lombard has moved beyond static demographic-based pricing to dynamic, usage-based models that reward low-mileage and safe drivers.

A. "Pay as You Drive" (Distance-Based) Logic

- **The Kilometer Slab Logic:** Policyholders choose a distance limit (e.g., 2,500km, 5,000km, or 7,500km). The premium is significantly lower than a standard "unlimited" policy because the risk exposure is mathematically lower.
- **Tracking & Top-ups:** Unlike telematics that require hardware, ICICI Lombard often uses a "Self-Declaration" model via the **IL TakeCare** app. Users upload an odometer reading at inception. If they exceed the slab, they can purchase "Top-up" kilometers to maintain Own Damage (OD) coverage.
- **Grace Period Logic:** If a user exceeds their slab and has an accident before topping up, the claim may be settled on a "Non-Standard" basis or rejected, though Third-Party (TP) liability remains intact.

B. "Pay How You Drive" (Behavioral) Logic

- **Safe Driving Score:** Using smartphone sensors (accelerometer/GPS) through the app, the insurer monitors:
    - **Braking & Acceleration:** Frequency of "Hard" events.
    - **Speeding:** Deviation from road-speed limits.
    - **Night Driving:** Higher risk weightage is assigned to driving between 12 AM and 5 AM.
- **Renewal Logic:** These scores are used to calculate an "NCB+" or a behavior-based discount at renewal, often ranging from 5% to 20%.

2. AI-Driven Claims: InstaSpect & Video Assessment

ICICI Lombard's "InstaSpect" feature uses mobile-first logic to eliminate the need for physical surveyors in minor accidents.

A. The Video Survey Algorithm

1. **Initiation:** The policyholder uses the app to start a live video stream with an ICICI Lombard claim manager.
2. **Damage Detection (AI):** The AI engine identifies the vehicle parts (bumper, headlights, panels) and assesses the severity of the dent or crack.
3. **Real-Time Estimation:** The system matches the detected damage against a standardized "Schedule of Rates" (SoR) for labor and parts.
4. **Instant Approval:** For claims under a specific threshold (typically ₹25,000–₹50,000), the insurer provides an "On-the-Spot" settlement offer. If the user accepts, the funds are transferred via IMPS/UPI immediately.

B. Total Loss & Salvage Logic

- **Constructive Total Loss (CTL):** Triggered when the cost of repair exceeds **75% of the IDV**.
- **Total Loss (TL):** Occurs in cases of theft or when the vehicle is beyond repair.
- **Return to Invoice (RTI) Logic:** A critical add-on that overrides IDV. In a TL/CTL scenario, the insurer pays the **last known invoice price** (including road tax and registration), essentially treating the car as "new" for the purpose of the payout.

3. Advanced Add-on & Exclusion Logic

ICICI Lombard provides several "Shield" add-ons that modify the standard indemnity math.

A. Zero Depreciation (Nil Dep) & Consumables

- **Zero Dep Logic:** Standard motor insurance follows a 5% to 50% depreciation schedule based on age and material (plastic/metal). The Zero Dep add-on settles the claim at **100% of the part cost**.
- **Limit Logic:** Most plans limit Zero Dep to **2 claims per year**.
- **Consumables Link:** While Zero Dep covers the *part*, it does not cover "Consumables" (grease, engine oil, brake oil). The **Consumables Cover** must be active to ensure these costs are not deducted from the claim.

B. Engine Protect (Hydrostatic Lock)

- **The "Consequential Loss" Clause:** Standard policies exclude engine damage if the driver attempts to start the vehicle in a flooded area (hydrostatic lock).
- **Protector Logic:** This add-on specifically covers the "Internal Child Parts" of the engine and gearbox, provided the damage is a direct result of water ingression or oil leakage due to an accident.

4. Administrative Mechanics: Break-in & NCB

- **Break-in Insurance Logic:** If a policy has lapsed for >90 days, the No Claim Bonus (NCB) resets to 0%. ICICI Lombard offers a "Mobile Self-Inspection" for break-in renewals, allowing the user to take a 360-degree video of the car to re-activate the policy without a physical visit.
- **NCB Protection:** This rider allows the policyholder to maintain their current NCB percentage (e.g., 50%) even if they make **one claim** during the policy year, preserving long-term premium discounts.

Summary of Logic Parameters

Feature	Logic Type	Operational Effect
InstaSpect	AI/Video Survey	Approval of minor claims in < 30 minutes.
Pay as You Drive	Kilometer Slabs	Lower premiums for low-usage vehicles.
CTL Threshold	75% of IDV	Determines the switch from "Repair" to "Total Loss."
Engine Protect	Exclusion Override	Covers hydrostatic lock (otherwise excluded).
NCB Retention	Claim Count Logic	Protects renewal discounts after one accident.